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Digital sovereignty as control: the regulation of digital finance in the European Union

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ABSTRACT

Crypto-finance is becoming increasingly popular, but the area is still unregulated in many ways. Its innovations promise to bring benefits by decreasing costs and increasing choices for intermediaries and customers. At the same time, and among other things, the breakdown of important exchanges and lawsuits against prominent figures illustrate potential challenges for investors, financial stability, and monetary sovereignty. How does the EU navigate the crypto verse? In this article, we analyse three regulations of the EU's Digital Finance Package through the lenses of discursive and historical institutionalism to understand when and where concerns about digital sovereignty motivate action. We argue that EU actors aim to regulate digital finance and retain sovereignty by employing different narratives that justify their policy choices. EU regulators upgrade existing regulations to cover digital finance areas that resemble traditional financial services and justify it by existing rationales on investor protection and financial stability. Digital finance poses security challenges that are new to financial regulation. EU actors employ a narrative of security threats to increase control in areas traditionally dealt with in cybersecurity.

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Introduction

Digital finance has expanded the means by which finance is undertaken, inciting EU policy-makers to adjust their regulatory and institutional frameworks. Examples include crypto-finance, starting with Bitcoin in 2008. The

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emergence of cryptocurrencies (digital representations of value) responded to bank collapses in the Great Financial Crisis. Based on distributed ledger technology (DLT), they promise to be disintermediated and unsupervised by public authorities. Similarly, powerful big tech companies such as Meta have considered creating their own so-called stablecoins (cryptocurrencies that are pegged to a commodity or currency) that could be used across a network of companies (Facebook Marketplace, Uber, Spotify). Stablecoins claim to eliminate problems like volatility and inefficiencies of international payments, among other things by guaranteeing convertibility with the US dollar or a basket of currencies (Morse, 2022). Some EU actors seek to tap crypto-finance as a source of investment and economic development, as expressed in the Commission's 2020 Digital Finance Strategy (European Commission, 2020a). They advocate a light-handed approach to regulation that supports innovation from a global, open, and private ecosystem.

Other EU actors see crypto as risks or even threats to consumers, investors, and financial stability due to asset price bubbles, a growing market size and subsequent market crashes bankrupting uninsured crypto-finance institutions and traditional financial services in the absence of sufficient regulation. Crypto-assets exhibited all of these risks, rising sharply in price after 2008 and then collapsing in 2022, growing larger in market size than the subprime mortgage instruments that unleashed the 2008 financial crisis, exhibiting enormous price swings (five times gold and four times US stock volatility between November 2021 and February 2022) (Financial Stability Board, 2022) and finally, bankrupting crypto exchanges and depositories due to fraud and mismanagement. In addition, EU regulators raised concerns about threats emanating from malicious attacks on cloud computing and other IT infrastructure, rather than the risks of crypto-assets themselves (Smith-Meyer, 2022). In its Digital Finance Strategy, the Commission expected these risks and threats, as well as anti-competitive behaviour, to increase as technology companies entered the financial ecosystem (European Commission, 2020a, p. 15).

EU institutions' response to these new emerging risks and threats was to develop a harmonised set of rules to bring the digital financial services industry under EU-level regulation and supervision. How can we explain the strengthened EU regulation in the area of digital finance? In the context of this special issue on EU's digital sovereignty (Falkner *et al*., 2023), we analyse three pieces of financial regulation from the EU's 2020 Digital Finance Package, adopted in May 2022 (DLT-Pilot Regime), December 2022 (Digital Operational Resilience Act, DORA) and May 2023 (Markets in Crypto-Assets Regulation, MiCA). We distinguish conceptually between changes in digital finance that are *upgrades* to pre-existing laws, policies and supervisors, and *innovations* that create new powers to provide new forms of control. Risks are associated with risks the EU knows how to handle, while perceived threats incite innovative digital regulations. We

also note the degree of legal obligation imposed on companies, from hard legal obligations and statutory supervision by regulators to (supervised) self-regulation in support of innovation.

To explain these policy changes, we apply historical and discursive institutionalist perspectives. We show that the EU justified its choices, on the one hand, with traditional narratives focused on financial stability and investor protection where it could perform upgrades of existing regulations, and on the other hand with new rationales focused on preventing security threats like cyber-attacks in financial services. MiCA upgrades existing financial market regulation and discourses to continue traditional control of financial market risks, from investor protection to financial stability, while the DLT pilot fosters market innovation in crypto-asset market infrastructures under the supervision of EU regulators focused on risk. Reference to digital sovereignty is used more strongly to justify new regulatory logics, as in countering cybersecurity threats for digital finance. Overall, the digital sovereignty narrative helps expand and cement legislative support for new moves. Our results contribute to the growing literature on the EU's rhetoric and action in terms of digital sovereignty and strategic autonomy (Falkner *et al*., 2023; Heidebrecht, 2023; Schmitz & Seidl, 2022), and literature on the evolution of European financial regulation (Donnelly, 2018; Högenauer *et al*., 2023; Posner & Quaglia, 2023; Ríos Camacho, 2021).

Digital sovereignty, digital finance and cybersecurity

Digital sovereignty has become a buzzword among EU policy-makers (Obendiek, 2021) and EU research scholars alike (Floridi, 2020; Pohle & Thiel, 2020). Scholarly work on digital sovereignty argued that although EU leaders such as Emmanuel Macron, Ursula von der Leyen or Angela Merkel have embraced the term, it 'lacks a clear definition and is deployed inconsistently' (Roberts *et al*., 2021, p. 2). Yet, as Lambach and Oppermann (2022) argue, the ambiguity of the concept of digital sovereignty, like that of strategic autonomy (Seidl & Schmitz, 2023), can serve as a 'coalition magnet' that enables heterogeneous actors to jointly promote a new policy agenda for different reasons.

In line with the introduction to this special issue, we identify claims that the EU needs to *control digital activity* and its different layers (like infrastructure, standards, content, or data, see (Chander & Sun, 2021; Floridi, 2020)) as the core purpose around which various actors layer different interpretations and policy initiatives. Control implies the ability to influence and restrict the manufacturing, design, use, and output of digital technologies with hard legal obligations and statutory regulation (Falkner *et al*., 2023).

This article investigates the relative importance of public policy debates and actions motivated by digital sovereignty, versus control of digital finance based on standard financial market regulation. An existing body of

literature addresses the implications of digital finance generally for public policy (Bernards & Campbell-Verduyn, 2019; Langley & Leyshon, 2021). Brown and Piroska (2022) argue that fintech pose risks to society and consumers by 'profiling' poor households into generators of financial assets (Gabor & Brooks, 2017). Others see the entrance of tech-driven private companies as a threat to the crucial payment sector (Brandl & Dieterich, 2023) and point to the relevance of digital finance in potential 'infrastructure war' (de Goede, 2020, p. 351).

However, literature on EU financial regulation (Donnelly, 2018; Högenauer *et al*., 2023; Ríos Camacho, 2021) has not yet dealt with crypto-based finance or cybersecurity aspects of financial regulation. It has focused on explaining the creation and functioning of new institutions and policies such as Banking Union (Howarth & Quaglia, 2016), Capital Markets Union (Braun *et al*., 2018), the de facto financial regulatory sovereignty of the EU when dealing with the United States (Posner & Quaglia, 2023) and the specifics of post-trade financial services after Brexit (Donnelly, 2023; Quaglia & Spendzharova, 2022). Given the EU's growing interest in identifying opportunities and threats of digital finance innovations, we need to understand how the EU expands regulatory and supervisory control over digital finance, which we theorise and empirically demonstrate below.

Digital sovereignty and digital control discourses: understanding regulatory motivations

The introduction to this special issue presents a framework to measure discursive and policy change in EU digital policy. Significant policy changes are defined as those that change policy substantially through important adjustments of instruments (like the level of fines), introduce new policies or even provide for new compelling reasons for regulation in the first place. Digital sovereignty can also be mapped by the discourse used to motivate these changes, which those institutions and policies embody (Pohle & Thiel, 2020, p. 9), and the way that discourse is used to create and adjust those policies and institutions over time. The authors define the bar for digital sovereignty as motivations that constitute a 'new *central positive reference point* in the policy discourse' (Falkner *et al*., 2023, pp. 5–7). These claims challenge the narrative of the 'free' internet (Mansell, 2011, p. 27), and finance outside of government-regulated institutions that crypto enthusiasts support.

We distinguish two major kinds of institutional change and associated shifts in policy narratives in digital finance with relevance to mapping and explaining whether and why actors strive for something that can be described as digital sovereignty. The first kind of institutional change is the *upgrade*. Upgrades are extensions and adaptations of existing rules,

regulations, authorities, and the paradigms that underpin their mission and mechanisms, relying on discursive continuity and generating incremental institutional and policy change. Control will be central where it previously existed without naming digital sovereignty. We can measure the degree of continuity or change not only in discourse, but in the strength of institutions, legal obligations, and supervision.

Upgrades are consistent with historical institutionalist accounts of policy and institutional development over time. Policy specialists and stakeholders replicate continuity of inherited rules, norms, values and practices as they grapple with new but related challenges that fit old templates, particularly if institutions have performed well in the past. While institutional failure would be expected to foster abrupt reforms (Hesse, 2021), upgrades are incremental changes to law and regulation responding to the evolution of already regulated activity (Capoccia & Kelemen, 2007; Pierson, 2000; Streeck & Thelen, 2005; Verdun, 2015).

The second kind of change is the *innovation*. Innovations are new institutions, legal rules, and motivations for introducing them. This can happen when old institutions break down, opening existing concepts to contestation and replacement, sometimes with advocacy coalitions pushing for particular choices over other alternatives. It can also happen when a new, exogenous sense of threat to another policy area is perceived as strong and action is urgent.

This sense of urgency for innovation is observable in cybersecurity. It carries over into legal innovations for cybersecurity in financial services, both traditional and crypto-based as well. As policy-makers link cybersecurity to digital finance, new discourse, law, policy, and institutions are established as new expressions of sovereign control in the face of an unprecedented threat. They correspondingly rely more intensively on processes familiar to discursive institutionalists (Schmidt, 2011): policy initiatives motivated by understanding the needs of public policy in the current moment.

Discursive institutionalism posits that actors exercise power through narratives that frame situations, justify responses, and persuade others to accept their views as a strategy (Carstensen & Schmidt, 2016, p. 323). EU actors regularly act as policy entrepreneurs, even on issues that are part of the intergovernmental pillar of EU competencies (Brandão & Camisão, 2021). They can use discursive devices strategically to promote their preferences, like invoking the concept of the market (Jabko, 2006), strategic autonomy (Schmitz & Seidl, 2022) and (digital) sovereignty (Bora, 2023, p. 1284) to build support. Discursive institutionalism analyses continuity as well as change but explains change better than historical institutionalism. When inherited institutions and norms do not speak to the risk or threat that lawmakers seek to control, references to new motivations are required to build and hold additional support. Vague concepts can also be used as coalition magnets to shore up political

support for novel regulations from non-specialist parliamentarians and governments. We therefore expect more explicit references to combatting threats to digital sovereignty in the field of cybersecurity than in financial regulation, which has robust institutions to build on. Innovations would be more likely than upgrades to analogue financial regulation.

A useful distinction within the respective discourses surrounding financial services versus cybersecurity is that between risks and threats, and links to concepts of appropriate public policy response that are central to discursive institutionalism. Risks are inherent to all activities, but central to finance and any market innovation. Risk-based regulation typically seeks to calibrate the degree of legal obligations and supervision to the probability and impact of risk and failure from market activity. Threats, however, imply vulnerability rather than simply risk, requiring state regulators to insist on higher standards, legal obligations, and regulatory powers, even where these need to be built from scratch, and/or intrude on other policy areas (Backman, 2023, p. 91). Another key difference is that threats imply persistent, malicious intent of adversaries that requires readiness and resilience, rather than risks of failure arising from accidents, market practice or regulatory design. While the EU's tradition to deter cyber threats goes also back to the 1980s (Carrapico & Farrand, 2020; Liebetrau, 2023), extending that threat to digital finance is a regulatory innovation.

The three regulations studied below differ in terms of their legal tradition and the objectives they aim to achieve. The MiCA and the DLT Pilot focus on the mitigation of traditional risks arising from new crypto operations, including monetary sovereignty and financial stability. By contrast, DORA is geared towards ensuring resilience of finance as critical infrastructure. We therefore expect that policy-makers build on traditional financial regulation and supervision aiming to reduce risks of crypto-finance when dealing with MiCA at the DLT Pilot. The options range from traditional statutory supervision (MiCA) where risk perceptions are strong and parallels between digital and regular finance are clear, to supervised self-regulation where greater uncertainty prevails about regulatory feasibility (DLT Pilot). The Pilot permits market participants to prove that they can create private infrastructure to manage traditional market and operational risks taking place in the cryptoverse. We expect that references to digital sovereignty are more likely to be visible when creating a new link between cybersecurity as threat prevention and the resilience of digital finance. This will imply significant policy changes to meet new perceived threats.

Empirical strategy

We conduct a theory-guided congruence analysis. The approach can be described as the drawing of inferences from the (non-)congruence of concrete observations with specific theoretical expectations from abstract theories for understanding the cases under study (Blatter & Blume, 2008,

p. 325). We demonstrate policy change and trace the justifications that accompanied these changes by demonstrating the process of changes over time (Bennet & George, 2005). Thus, we first describe the substantive content of the policy area addressed and the status quo ante (before the adoption of the EU's digital finance package). Secondly, we examine important regulatory elements like the introduction of harmonised legal obligations and EU agencies with strong regulatory powers and vertical coordination over national competent authorities and the positions of important policy actors on these elements. Third, we elaborate on how different actors promoted these changes by evoking different discursive strategies that ultimately enabled policy changes. This allows us finally to test for congruence with our theoretically derived expectations and to situate our findings in the context of this special issues framework on the relationship of discursive and policy changes towards digital sovereignty.

We selected influential cases of EU regulation of digital finance. While influential cases are typically not representative of a larger sample of EU digital financial regulation we choose influential cases because we expect that they can show more significant empirical effects (Seawright & Gerring, 2008), which allows us to draw conclusions on the development of EU digital finance regulation more generally. We examine the three policy-sub-areas addressed by the regulations of the EU's Digital Finance Package, issued on 24 September 2020 by the Commission. The package contains three following three regulations: MiCA (Regulation (EU) 2023/1114), DLT-Pilot (Regulation (EU) 2022/858), and DORA (Regulation (EU) 2022/858). All three regulations have the Single Market Article 114 as their legal basis, while they differ substantially in terms of regulatory approach and content (see below).

In order to maximise observation points (Geddes, 2003), we use a snowball approach starting with official legislative and position papers to identify relevant additional data in the form of statements, speeches, position papers, press releases of important actors: the European Supervisory Authorities (ESAs), the European Central Bank (ECB), the European Commission, Council, Parliament and industry players. We also analysed the parliamentary debates about the three regulations (available on the EUR-Lex website) and statements of leading Commissioners, like Margrethe Vestager, responsible for making 'Europe Fit for the Digital Age'. We further conducted six topic-centred interviews for triangulation purposes. We assured all interviewees strict confidentiality so that we provide only anonymised information about the interviews in Appendix 1.

Upgrades in EU digital finance regulation: crypto-assets and market infrastructure

EU regulation of crypto-assets and infrastructure is primarily contained in the MiCA Regulation adopted in May 2023. It effectively upgrades previous

regulations designed to ensure regulatory control of financial markets with financial stability and/or investor protection in mind and applies it to crypto-assets. MiCA's most prominent predecessor is the Market in Financial Instruments Directive (MiFID) II of 2014, which upgrades and consolidates multiple directives governing actors and activity in securities markets with financial stability and investor protection risks in mind. MiCA also addresses risk-oriented standards for post-trade financial services that upgrade provisions of the European Markets Infrastructure Regulation (EMIR) for traditional financial services companies. Parallel to MiCA is a pilot project on whether market participants can manage risks of trading and clearing of crypto-assets that EU law defines as securities, based on supervised self-regulation of market participants, rather than statutory regulation.

MiCA largely reflects historical institutionalist expectations. EU actors' narratives based on innovation coupled with risk reduction and financial stability show discursive continuity in EU financial market regulation. Occasional reference to monetary sovereignty, which goes beyond the traditional focus on financial markets, is used additionally as a coalition magnet to build majorities rather than design regulation. MiCA represents a comprehensive, although incremental, change in EU financial market regulation given its hard legal obligations and strong supervision, reflecting existing financial market law. Thus, the EU maintains itself as a rule-maker and sovereign in financial market regulation by implementing upgrades related to crypto-based finance without focusing on digital sovereignty explicitly.

Legislating the markets in crypto-assets regulation

The 2018 Commission's FinTech Action Plan contained its definition of the main risks and challenges of crypto-assets, paralleling assessments by the Financial Stability Board (FSB). It noted that existing regulations failed to cover crypto-assets sufficiently. Crypto-assets were subject to high volatility, fraud, operational weaknesses, and crypto-asset exchange vulnerabilities (European Commission, 2018, p. 3) that might pose risks to financial stability (Financial Stability Board, 2020, p. 1; 2022). The European Supervisory Authorities in financial services (ESAs), especially the European Banking Authority (EBA), played a key role in supporting the Commission's drive to standardise the definition of crypto-assets across EU Member States and the identification of their risks and opportunities given their technological novelty. The ESAs started warning about the risks of buying or holding crypto-assets in 2014 (European Banking Authority, 2014, p. 46), and supported the drive to regulate in MiCA (European Supervisory Authorities, 2022).

Following the ESAs' advice, the Commission proposal of 24 September 2020 (European Commission, 2020d) considered that most crypto-assets were not securities, and therefore would fall outside existing EU legislation

focused on minimising risks to consumers, investors and the market. The Commission stressed that the EU needed to extend existing securities and macroprudential regulation to 'instil appropriate levels of consumer and investor protection and market integrity […]' and 'safeguards to address potential risks to financial stability and orderly monetary policy' (European Commission, 2020d, pp. 2–3). Historical arguments of what to regulate and not prevailed over ideas about digital sovereignty.

MiCA also adopted inherited capital markets discourse that poor regulation and divergent national rules hurt innovation and competition, preventing the proper functioning of the Single Market. In other words, innovation and regulation go together. In her 2019 mission letter to Commissioner Dombrovskis, Commission President Ursula von der Leyen addressed not only risks crypto-assets may pose but called for 'a common approach with Member States on cryptocurrencies […] to make the most of the opportunities they create … ' (von der Leyen, 2019, p. 6). A 2019 joint declaration of the Council and the Commission concurred, stating that 'technological innovation can produce great economic benefits for the financial sector' (Council of the European Union, 2019). Market actors urged an innovation-friendly approach to EU digital financial regulation that was technology-neutral, innovation-friendly, and reduced fragmentation of the Single Market for digital financial services (Association for Financial Markets in Europe, 2020a; International Association for Trusted Blockchain Applications, 2021).

Meanwhile, the Council reiterated in 2021 that the lack of an EU-wide regulatory framework for crypto-assets could generate arbitrage and an unwelcome regulatory race to the bottom, creating risks of market failure. It therefore recommended that the EBA and the European Securities and Markets Authority (ESMA) address the classification of crypto-assets, while the ECB should provide the National Competent Authorities (NCAs) with an opinion on the risks for payment systems and monetary sovereignty (Council of the European Union, 2021b, p. 22).

In addition to these established arguments in the single market debate, MiCA represented an effort to alleviate the consequences that the rise and widespread adoption of 'global stablecoins' may generate in the European market (Interview VI). Stablecoins are essentially crypto-assets that are pegged to a commodity or currency. In 2019, Facebook announced its plans to issue its own stablecoin through the Diem Association (formerly the Libra Association). Given Facebook's large number of users, its stablecoin could lead to the global adoption of stablecoins. This caused concerns among EU legislators and supervisors about the EU's monetary sovereignty.

Both the Council and the Commission agreed that once a 'stablecoin' reached significant scale, it could create 'new potential risks to monetary sovereignty, monetary policy, the safety and efficiency of payment systems, financial stability, and fair competition' (Council of the European Union,

2019). Both institutions underlined the necessity to safeguard monetary sovereignty (ensuring the euro remained the primary currency) if global stablecoins became increasingly accepted (European Commission, 2020a, p. 9)(Council of the European Union, 2021b, p. 5). Within Parliament, debates on the Digital Finance Strategy referenced challenges to monetary sovereignty from stablecoins (European Parliament, 2020a), while its 2020 resolution on digital finance addressed issues of financial stability and effective monetary policy, without referring to (monetary) sovereignty (European Parliament, 2020b). Supervisors, like the European Insurance and Occupational Pensions Authority, underlined in a 2021 discussion paper that Libra could interfere with monetary policy and monetary sovereignty if it reached a significant scale and a large volume of domestic payments (European Insurance and Occupational Pensions Authority, 2021).

The adopted MiCA was a clear move towards more regulatory control of digital finance based on well-known risks. It envisaged the first EU-wide regulatory framework for crypto-assets, the issuers, and crypto-asset service providers (CASPs), thereby for the first time ensuring regulatory coverage of crypto-assets based on traditional regulation. MiCA created strict requirements for issuers of crypto-assets in Europe and CASPs. CASPs comprise trading venues or crypto exchanges, and custodian wallet providers (MiCA-R., Art. 3 (16)). Examples include that all issuers of crypto-assets have to draft 'crypto-asset white papers', that include disclosure, conduct, and liability rules (e.g., MiCA-R, Rec. 24), and then benefit from a European passport for their products (permission to solicit investment throughout the Single Market). All CASPs are subject to strict minimum capital requirements (MiCA-R, Art. 67), and must comply with governance and conduct requirements (MiCA-R, Art. 68). Paralleling provisions on central counterparties under EMIR, CASPs must cover operational risks by protecting users' wallets assuming liability if they lose investors' crypto-assets (MiCA-R, Art. 70).

MiCA classified crypto-assets into three main categories: Crypto-assets and utility tokens, asset-referenced tokens (ARTs), and e-money tokens, and defines respective regulatory and supervisory obligations 'depending on the risks they entail' (MiCA-R, Rec. 18). According to MiCA, ARTs and e-money tokens can pose challenges for financial stability, monetary policy transmission or monetary sovereignty (e.g., MiCA-R, Rec. 59). Such 'stablecoins' reference, other assets including currencies to maximise stability, whereas an e-money tokens reference one single fiat currency. As they can be used by many holders and replace the euro as means of payment, their use could raise specific challenges: financial stability, monetary policy transmission or monetary sovereignty.

To meet these risks, MiCA enshrined stronger regulatory oversight for ARTs and e-money tokens from issuers outside the EU or referencing currencies than the euro (e.g., MiCA-R., Rec. 45, 104). Issuers of ARTs and CASPs are

further obliged to have a registered office in the EU to guarantee their proper supervision (MiCA-R, Rec. 42, 74). Issuers of e-money tokens must be authorised as a credit institution or as an e-money institution under the E-Money Directive (Directive 2009/110/EC). In other words, MiCA upgrades existing EU regulation to control the crypto-asset market without stifling innovation already allowed. The Commission, Council and EP agreed on the need to ensure that stablecoins remain subject to monitoring and control for traditional purposes. They shared the concern that once ARTs became 'significant' (Council of the European Union, 2021b; European Commission, 2020d; European Parliament, 2022), they could be used as a means of exchange and to make large volumes of payment transactions, which could threaten monetary sovereignty. As a response, the EBA will supervise all 'significant' stablecoins (MiCA-R, Art. 43 (7); Art. 56 (6)) that fulfil a number of quantitative criteria, such as when 10 million or more entities hold the ART or its aggregate value (market capitalisation) is higher than EUR 5 million (MiCA-R, Art. 43).

Conversely, MiCA treats e-money tokens as electronic money (MiCA-R, Art. 48 (2)), so that they become subject to national competent authority supervision under existing law (E-Money Directive), unless they are 'significant' and thus supervised by the EBA. Because ARTs are not regulated by the E-Money Directive, and have broader applicability, as in the case of Libra, they are seen to pose greater potential risks to the smooth operation of payment systems using the euro, monetary policy transmission and monetary sovereignty. The Parliament and the Council accordingly supported the proposal that the ECB have authorisation powers over issuers of significant ARTs (MiCA-R., Art. 21(4)) (Council of the European Union, 2021b, p. 117; European Parliament, 2022, p. 51).

MiCA's regulatory structure therefore resembles traditional financial supervision, with EU oversight reserved for the riskiest entities and activities, and national authorities doing the rest. In this regard, national authorities will be responsible for carrying out the supervisory functions and duties provided for in MiCA for ensuring enforcement (MiCA-R, Art. 93). This parallels traditional financial supervision in the Single Supervisory Mechanism (SSM), where supervisory functions for the largest and riskiest entities fall to the ECB (Högenauer *et al*., 2023). MiCA parallels the distribution of tasks between the member state and the supranational level based on size and risk. The EBA supervises the most 'significant' stablecoins that pose risks to the market, the functioning of the euro, and its monetary sovereignty. CASPs that have more than 15 million active users are classified as 'significant crypto-assets service providers' and will be supervised by the national authorities (MiCA-R, Art. 85). However, the European Securities and Markets Authority (ESMA) will have an 'intervention power' to ban or limit the provision of crypto-asset services by the CASPs in case of threats to market integrity, investor protection or financial stability (MiCA-R, Art. 103).

In line with our theoretical expectations, MiCA replicates the EU's approach to balancing the potential benefits and risks of financial markets, upgraded to cover the use and industry activities of crypto-assets. EU actors' narratives based on innovation coupled with risk reduction and financial stability show discursive continuity with EU financial market regulation. Its hard legal obligations build on existing regulations and powers. It also creates a supervisory framework that resembles that of post-2008 financial regulation in Europe, dividing supervision between EU authorities (ECB and ESAs) for the most significant entities, and national authorities for the less significant ones.

The DLT Pilot in post-trade services

In addition to promoting investor interests and financial stability through transparency, oversight and authorisation, MiCA also attempts to address provisions provided by post-trade services designed to ensure financial stability. The European Market Infrastructure Regulation (EMIR) of 2012 mandated central counterparties (CCPs), central security depositories (CSDs) and trade repositories for pre-crypto financial market transactions. CCPs prevent market collapse by ensuring that parties to contracts get paid, CSDs provide deposit facilities, and trade repositories register data on ownership and transactions that large trading markets require to function. The description of CASPs in MiCA parallels these functions, indicating an upgrade from EMIR.

The DLT Pilot allows private actors to construct rules and institutions for trading and post-trade services in crypto-assets under the guidance of EU supervisors, just as the 2012 EMIR provisions did. This continues a belief that market participants are best suited to designing and operating infrastructure within limits designed to contain the risk of system failure. CASPs, like clearing houses, central counterparties, and trade repositories under EMIR, are provisionally allowed to retain ownership and control of the process to promote innovation and expertise, while supervisors seek to ensure that procedures, standards, and mechanisms for addressing payment and insolvency problems reasonably contain financial stability risks.

The Commission's proposal underscored the need for suitable market infrastructures for buying and selling crypto-assets in both primary financial markets (where new instruments would be bought) and secondary markets that ensure investor, consumer and market safety. Without this, crypto-asset markets could not be expected to develop. At the same time, there were concerns that private activity could circumvent market infrastructure regulations (EMIR) mandating features to guarantee payments (central counterparties), deposits (CSD requirements) and reporting applicable to traditional financial service providers (European Commission, 2020b). Safe innovation replicating traditional measures was the goal.

As the guardian of monetary and systemic financial stability in the euro area, the ECB's stance on the pilot project was of particular importance. It supported the DLT Pilot as a means of fostering innovation with potential efficiency benefits in the financial system, particularly for cross-border payments. However, it also supported limitations that would help contain the financial stability risks of the experiment. The ECB encouraged new regulatory conditions on the original issuers of the securities to ensure that DLT-platforms would not serve as a magnet for poor-quality assets (junk) that could create bubbles and unleash crashes (European Central Bank, 2021, p. 6). It also foresaw potentially problematic compliance with the Capital Requirements Directive (Directive 2013/36/EU) on reserve holdings without a centralised party involved.

The ECB saw it as essential for financial stability that EU licensing and capital requirements rules for credit institutions be updated to include DLT operators rather than relaxed. Financial market infrastructures would also need to find a way to minimise and control credit and liquidity risks (European Central Bank, 2021, p. 7). The ECB also underscored that the pilot should address investor protection in case of bankruptcy and the removal of assets from trading (European Central Bank, 2021, p. 9). Finally, it urged analysis of potential risks to monetary policy arising from increased payment settlement using bank money and e-tokens without using central bank money (European Central Bank, 2021, p. 3). Overall, the ECB promoted the upgrading of existing rules based on established narratives around financial regulation.

The pilot legislation passed through Parliament and Council in one reading. The Committee on Economic and Monetary Affairs underlined that the pilot should not circumvent or undermine existing supervisory control and regulation of traditional market infrastructures, but that it should also allow a wider range of instruments (including sovereign bonds) and institutional investors (European Parliament, 2021a). The Council accepted the proposal with the Parliament's observations without any need for changes.

The DLT Pilot's agreed rules extend pre-digital supervision to crypto by placing selected multilateral trading facilities (exchanges) and securities settlement systems under the supervision of ESMA and national counterparts (DLT-Pilot R, Art. 12). Operators need to be registered and authorised for the purposes of existing regulation, particularly the Market in Financial Instruments Directive (MiFID) covering transparency and investor protection; EMIR; and the Central Securities Depositories Regulation (CSDR, Regulation EU/909/2014), which sets out the minimum capital standards and operating requirements for central security depositories for traditional financial instruments. Whereas central security depositories would normally be required to register the transfer of securities on a central platform, the pilot allows these authorised companies to allow the transfer of securities in a

decentralised manner using DLT instead. This simplifies procedures and contains transaction costs while still keeping track of who owns the securities.

To limit the scope of the pilot, crypto-assets that are categorised as commodities or derived from them are not eligible for trade (DLT-Pilot R, Art. 3). This excludes physical commodity futures (tokens representing raw materials, for example) and non-fungible tokens. Instead, assets must be eligible securities, i.e., based on a company's business activities, paralleling traditional securities law. However, the pilot is also limited to securities that are not very liquid, meaning there is not much trading in them. This contains potential risks if the pilot fails. Central securities depository operators will not have to register DLT transactions on their own books, relying on decentralised records instead. Part of the pilot is figuring out how to ensure that decentralised ownership, trading, custody and reporting are as safe and secure as traditional trading and post-trade services regulated by EMIR. In addition to financial stability, the DLT Pilot also mandates cybersecurity for digital infrastructure, as set out in DORA (below).

Another restriction on the pilot mirroring concerns of the ECB is a preference for settlement in central bank or bank money over payment in crypto-assets. Settlement in crypto-coins and e-money tokens as defined and regulated in MiCA remains possible however, since settlement in traditional money as the ECB prefers collides with the ambition of DLT-based crypto transactions. To hedge against problems, traditional supervisors (ESMA and NCAs) have the authority to revoke authorisation, while market infrastructure companies have a duty of cooperation with competent authorities (DLT-Pilot R, Art. 11).

Overall, the DLT pilot opens the door to meaningful experimentation with crypto-based market infrastructure. However, concerns of the Commission, of the ECB, of ESMA and of the European Parliament about traditional risks to the integrity and safety of financial markets remain strong. MiCA replicates much of EMIR, and the Pilot mirrors, references and upgrades much of the EU's recent advances in reclaiming regulatory control over financial markets, together with market participants. The Pilot promotes innovation through supervised trials and takes existing supervisors and regulations as a baseline and subjects companies to these in the interest of financial stability, as well as investor and consumer protection.

Thus, we observe a significant policy upgrade that aims to control crypto-assets and market infrastructures rather than allowing them to escape public oversight and regulatory control for the pure sake of innovation. These arguments suffice, without referencing digital sovereignty directly. The substance is one of strengthened regulatory control, but digital sovereignty is not a central reference point in the discourse of the most relevant actors. It is just one of the reference points among others such as regulatory fragmentation, protection of consumers and investors, market integrity, and financial stability.

Cybersecurity for finance: digital sovereignty in the operational resilience act

Before the EU's digital finance package, the consideration of cyber threats was not part of financial regulation. By contrast, in DORA, EU policy-makers stressed that 'increased digitalisation and interconnectedness also amplify ICT risk, making society as a whole, and the financial system in particular, more vulnerable to cyber threats or ICT disruptions' (DORA-R, Rec. 1). DORA brings recent developments from the area of cybersecurity to financial regulation. The 2019 Cybersecurity Act (Regulation (EU) 2019/881), crystallized the EU's regulation and governance of cybersecurity into a new area of robust legal requirements and institutional powers. It supplanted the loose coordination of national authorities and laws with a harmonised regime entailing both vertical and horizontal coordination between the EU Agency for Cybersecurity (ENISA), national counterparts, and actors like the more than 700 Computer Security Incident Response Teams. The Cybersecurity Act also provides the definition of cyber threats (Cybersecurity Act, Art. 2 (8)), namely as 'any potential circumstance, event or action that could damage, disrupt or otherwise adversely impact network and information systems, the users of such systems and other person'. With DORA, concerns about such cyber threats enter the policy discourse of digital financial regulation.

DORA focuses on cybersecurity in the cryptoverse rather than financial transactions and risks. While the EU's post-2008 financial reforms led to the creation of single rulebooks in banking and securities to regulate financial risks linked to financial services, they did not cover digital operational resilience and, therefore, did not address the risks that arise from digitalisation. Neither the EU's first cybersecurity law, the Network and Information Security (NIS) Directive of 2016 (Directive (EU) 2016/1148), nor the Cybersecurity Act addressed financial services directly. However, the EU addressed this gap in part through the NIS 2 Directive (Directive (EU) 2022/2555) by including more detailed provisions and supervision for financial services, and more specifically through DORA. DORA requires financial entities to have a sound, comprehensive and well-documented ICT risk management framework as part of their overall risk management system (DORA-R, Art. 6 (1)) in response to potential threats to the EU's critical infrastructure, which includes financial services.

Some elements in the policy discourse about DORA resembled the well-known logic in the context of Single Market legislation and have Article 114 as its legal basis. The Commission stressed the need for harmonisation, without which the Single Market freedoms for financial services could be hampered (European Commission, 2020c, p. 2). Financial industry associations accepted that the Single Market could benefit from harmonised ICT security governance arrangements to prevent threats like fraud, loss or

theft (Association for Financial Markets in Europe, 2020b). The Council's negotiating position concurred and stated that the harmonisation of an ICT risk framework through DORA would contribute to 'the smooth functioning of the single market' (Council of the European Union, 2021c, p. 7). The Parliament's rapporteur, Billy Kelleher (Renew, Ireland), advocated that DORA's provisions should not hamper the competitiveness and attractiveness of the EU with requirements not required elsewhere (European Parliament, 2021b, p. 83). For him, the primary objective was to ensure that 'the EU must become a key player on digital finance' (European Parliament, 2021b).

While references to established arguments in terms of innovation and financial stability of the financial services were present, we observed more security-related arguments in the DORA policy discourse. These stress elements like international challenges, systemic risks that arise because of dependencies on a few central counterparties (which is mostly referring to big technology companies that are based outside of the EU), and cyber threats (European Parliament, 2021b).

The desire to defend sovereignty against foreign actors is also addressed in concerns about systemic risks attached to big tech cloud services. In line with the Commission's proposal, the Council and the EP argued that 'the broader issue of counteracting systemic risk which may be triggered by the financial sector's exposure to a limited number of critical ICT third-party service providers is barely addressed by Union legislation' (Council of the European Union, 2022b). Such critical ICT providers, a few important big technology companies, offer services like cloud computing infrastructures on which many digital financial services rely. Actors like the ECB, which support the development of digital finance in the EU (European Central Bank, 2020, p. 2), were also concerned about concentration. '[W]hen EU financial service providers increasingly rely on (critical) services offered to them by non-EU technology firms, EU *sovereignty* with regard to those services (e.g., the ability to regulate and supervise those non-EU service providers) decreases' (European Central Bank, 2020, p. 28, italics by authors).

While systemic risks, though in a very different fashion, were part of the traditional financial regulatory discourse, the very prominent rhetoric relating to defence and threats around DORA is new to financial regulation. For example, the Commission justified its proposal by highlighting the issue of cyber threats, and aimed to promote 'financial entities' defensive capabilities and threat detection techniques (European Commission, 2020c, p. 11). The rhetoric of threats was widely shared, particularly by the Council, which justified its position to ensure that all firms 'can withstand all types of ICT-related disruptions and threats, in order to prevent and mitigate cyber threats' (Council of the European Union, 2021a, 2022a).

The ESAs were concerned early about the risks that stem from non-harmonisation of rules and ICT dependencies of EU firms, including those with

foreign technology firms. The EBA called for an EU-wide harmonised framework (European Banking Authority, 2018, p. 25) and argued that 'cybersecurity measures need to be applied throughout the financial system, in particular where multiple entities interconnect' (European Banking Authority, 2018, p. 23). The cross-cutting nature of ICT risks and the necessity for a harmonised EU legal framework was emphasised in a joint technical advice in 2019 in which all three ESAs argued that there was a lack of 'explicit references to ICT and cybersecurity risk' […] so that 'every relevant entity should be subject to general requirements on governance of ICT, including cybersecurity, to ensure safe provision of regulated services' (European Supervisory Authorities, 2019, p. 4).

DORA covers almost the entire financial sector, including CASPs, and five important areas for digital operational resilience: ICT risk management, incident reporting, testing, third-party risk, and information sharing (DORA-R, Art. 2). DORA's supervisory structure widely parallels the supervisory structures in traditional financial regulation, so that the competent authorities that supervise entities in accordance with existing financial regulations also become responsible for supervising the operational resilience provisions of DORA (DORA-R, Art. 46). For example, most credit institutions are supervised by the national competent authorities (NCAs) and only by the ECB where they are classified as significantly important entities in accordance with the SSM Regulation (DORA, Art. 46 (a)).

The question of coordination and cooperation was crucial. The ESAs together with the EU Cybersecurity Agency ENISA develop common draft regulatory technical standards for further harmonisation of ICT risk management tools, methods, processes, and policies (DORA-R, Art. 15). In addition to the NCAs, the ESAs and ENISA cooperate with the ECB as the supervisor of large banks in the euro area. Because the system is widely decentralised, DORA tasks the ESAs with preparing, in consultation with the ECB and ENISA, a joint report exploring the feasibility of setting up a single EU Hub (DORA-R, Art. 21). By centralising information, such an EU Hub should improve cooperation against cyber incidents and threats.

In addition to this, a key novelty of DORA is to bring critical ICT providers into financial supervision. For this purpose, the ESA that is responsible for overseeing most of the financial institutions that use the services of the critical ICT provider will be designated 'lead overseer' (DORA-R, Art. 31(1)). Lead overseers will assess whether critical providers have sufficient arrangements to manage ICT risks (DORA-R, Art. 33) and significant powers to ensure supervision, including the conduct of general investigations and inspections in and outside of the EU (DORA-R, Art. 35-36). The lead overseer may impose a periodic penalty payment to coerce the critical providers (DORA-R, Art. 50-51).

DORA is a case of innovation in financial market regulation based on new responsibilities and driven by perception of threat, linked to concerns about

finance as critical infrastructure. Innovations in DORA are reflected in both EU actors' discourse and the final policy and institutional changes. In line with our expectations, we observe some discursive discontinuity in financial market regulation because ICT risks and security threats are now considered part of financial market regulation in Europe. There is no significant reference directly to digital sovereignty. However, DORA needs to be seen in a policy discussion involving all the major decision-making and regulatory institutions that seek to ensure that digital financial services are regulated from inside the EU, and prepared to meet cybersecurity attacks. They promoted policy choices by referring to narratives that are new to financial regulative discourse and that highlight in many respects threats more than risks and originate from the tradition of cybersecurity (and not financial regulation).

DORA also constitutes a significant policy change for cyber security and financial market regulation. While cyber security elements are new to financial regulation, DORA also entails elements that are new for cybersecurity. The 2019 Cybersecurity Act (and the NIS 1 and 2 Directives) does not address the issue of critical third-party service providers and provisions for their supranational supervision through ENISA. By contrast, this is the case in DORA which establishes for the first time these powers for the ESAs by designating them as lead overseers of critical providers.

Discussion and conclusion

This paper examined three legislative components of the EU's Digital Finance Package to shed light on whether digital sovereignty is a motivating factor in how Europe views and regulates digital finance, with a focus on crypto-assets. Rather than undertake a narrow search for overt narratives of digital sovereignty, we examined concrete moves towards strengthened regulatory control. This constituted whether EU institutions were intent on stronger regulating digital finance and crypto-assets in particular, why they did so, and how they designed those policies and institutions. In this way, we could speak to the special issue's framework of determining the direction and degree of moves to extend European control over the digital world, even in the absence of specific references to digital sovereignty. We covered rules governing the issuance and trading of crypto-assets (the Markets in Crypto-Assets Directive), on providing market infrastructure to ensure that trading could take place safely (the DLT Pilot), and for ensuring that all these activities would have high cybersecurity provisions (Digital Operational Resilience Act).

Our findings show two things. First, EU actors in the financial services sector exist in an environment where most issues, institutions, laws and paradigms about what the EU is there to do and the challenges it faces are inherited from existing financial market regulation. Control is central, problems

and solutions are well-understood as risks that can be managed, and digital regulation is modelled on upgrades of pre-digital versions without an explicit reference to digital sovereignty. The EU extends its control, and one could say digital sovereignty, but at best, references to digital sovereignty are muted, and at best coalition magnets to garner legislative support from political generalists, having no impact on the legislation itself. The form and content of EU law and regulation is firmly built on historical principles, institutions and legislation. EU actors intend to be as strong in the digital realm as thy are in the traditional one, aligning digital control with traditional regulation of financial markets.

Second, EU actors advocating for cybersecurity in digital finance exist in an environment where most issues, institutions, laws and paradigms about what the EU is there to do and the challenges it faces are new and different from anything the financial sector has confronted before. Regulatory control is again central, but accompanied by a sense of threat and malicious intent that requires urgent, mandatory and robust preparation in an area unfamiliar to financial services companies and regulators, and that requires cross-overs between cybersecurity and financial services regulators. This makes financial services more robust and resilient by mandating and mainstreaming cybersecurity measures into the activity of both European companies and supervisory authorities. These innovations took place due to discursive narratives around response to threats that rallied support for this new extension of EU control over cybersecurity in financial services as critical infrastructure. This combination continues to the present day in the European Commission's (2023) newly released Economic Security Strategy, which goes even further to identify financial services as critical infrastructure, and protection of critical infrastructure from inter alia cybersecurity attacks as a matter of national security. This is a novum for the EU. Even without this last move by the Commission, the three regulations studied here constitute the first attempt globally to comprehensively regulate crypto-assets and set-up a supervisory structure for the cryptoverse, so that risks are regulated and supervised as strongly as in the traditional realm.

Within the debate about digital sovereignty, and within this special issue we show significant advances in regulatory control of digital finance, particularly of crypto markets. The changes are simultaneously strong and inconspicuous, extending control without hammering on digital sovereignty as a motivator. We see the EU continue to permit crypto market participants to pursue innovation and provide market infrastructure along pre-crypto patterns; we see sweeping new responsibilities for crypto companies and supervisors; and we see innovative new links between financial regulation and cybersecurity designed to marry innovation, risk management and security in ways found nowhere else. Thus, our results also indicate an intrusion of a security-related discourse to other EU policy areas. In this regard, our

results suggest the increasingly important security discourse (Backman, 2023; Carrapico & Farrand, 2020) will gain track in EU financial regulation, with longer term consequences for the EU.

From here, our findings point to questions for further research about how effective these regulatory incursions into the cryptoverse will prove to be in practice, and to what extent EU regulations spread to the United States and other significant markets (Bradford, 2020). This follows since the EU effectively is trying to regulate not only EU companies, but others like American cloud service providers (Brandl & Dieterich, 2023; Langley & Leyshon, 2021).

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No potential conflict of interest was reported by the author(s).

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References

Association for Financial Markets in Europe. (2020a). Consultation response. European commission public consultation – An EU Framework for Markets in Crypto-Assets.

Association for Financial Markets in Europe. (2020b). Consultation Response. European Commission Public Consultation on 'Digital Operational Resilience: Making the EU financial sector more secure'. https://bit.ly/48CZmjD.

Backman, S. (2023). Risk vs. threat-based cybersecurity: the case of the EU. *European Security*, *32*(1), 85–103. https://doi.org/10.1080/09662839.2022.2069464

Bennet, A., & George, A. (2005). *Case Studies and Theory Development in the Social Sciences*. MIT Press

Bernards, N., & Campbell-Verduyn, M. (2019). Understanding technological change in global finance through infrastructures. *Review of International Political Economy*, *26* (5), 773–789. https://doi.org/10.1080/09692290.2019.1625420

Blatter, J., & Blume, T. (2008). In search of co-variance, causal mechanisms or congruence? towards a plural understanding of case studies. *Swiss Political Science Review*, *14*(2), 315–356. https://doi.org/10.1002/j.1662-6370.2008.tb00105.x

Bora, S. I. (2023). 'A sovereign Europe'? Strategic use of discourse at the service of french economic interests in EU politics (2017–2022). *JCMS: Journal of Common Market Studies*, *61*(5), 1281–1297. https://doi.org/10.1111/jcms.13463

Bradford, A. (2020). *The Brussels effect: How the European Union rules the world*. https://doi.org/10.1093/oso/9780190088583.001.0001.

Brandão, A. P., & Camisão, I. (2021). Playing the market card: The commission's strategy to shape EU cybersecurity policy. *JCMS: Journal of Common Market Studies*, *60*(5), 1335–1355. https://doi.org/10.1111/jcms.13158

Brandl, B., & Dieterich, L. (2023). The exclusive nature of global payments infrastructures: the significance of major banks and the role of tech-driven companies. *Review of International Political Economy*, *30*(2), 535–557. https://doi.org/10.1080/09692290.2021.2016470

Braun, B., Gabor, D., & Hübner, M. (2018). Governing through financial markets: Towards a critical political economy of Capital Markets Union. *Competition & Change*, *22*(2), 101–116. https://doi.org/10.1177/1024529418759476

Brown, E., & Piroska, D. (2022). Governing fintech and fintech as governance: The regulatory sandbox, riskwashing, and disruptive social classification. *New Political Economy*, *27*(1), 19–32. https://doi.org/10.1080/13563467.2021.1910645

Capoccia, G., & Kelemen, R. D. (2007). The study of critical junctures: Theory, narrative, and counterfactuals in historical institutionalism. *World Politics*, *59*(3), 341–369. https://doi.org/10.1017/S0043887100020852

Carrapico, H., & Farrand, B. (2020). Discursive continuity and change in the time of COVID-19: the case of EU cybersecurity policy. *Journal of European Integration*, *42*(8), 1111–1126. https://doi.org/10.1080/07036337.2020.1853122

Carstensen, M. B., & Schmidt, V. A. (2016). Power through, over and in ideas: conceptualizing ideational power in discursive institutionalism. *Journal of European Public Policy*, *23*(3), 318. https://doi.org/10.1080/13501763.2015.1115534

Chander, A., & Sun, H. (2021). Sovereignty 2.0. Georgetown Law Faculty Publications and Other Works(2404).

Council of the European Union. (2019). Joint statement by the Council and the Commission on "stablecoins" [Press release].

Council of the European Union. (2021a). Digital finance package: Council reaches agreement on MiCA and DORA [Press release]. https://bit.ly/46frvLU.

Council of the European Union. (2021b). *Interinstitutional File from the General Secretariat of the Council to Permanent Representatives Committee. Proposal for a Regulation of the EP and of the Council on MiCA – Mandate for negotiations with the EP*. (EF 351. ECOFIN 1105. CODEC 1493).

Council of the European Union. (2021c). *Interinstitutional File. Proposal for a Regulation of the European Parliament and of the Council on digital operational resilience for the financial sector*. (2020/0266 (COD)). https://bit.ly/48JSFw3.

Council of the European Union. (2022a). Digital finance: Council adopts Digital Operational Resilience Act [Press release]. https://bit.ly/45lPUOz.

Council of the European Union. (2022b). *Final Compromise Text on the Proposal for a Regulation on digital operational resilience for the financial sector*. https://bit.ly/3FqWtVP.

de Goede, M. (2020). Finance/security infrastructures. *Review of International Political Economy*, *28*(2), 351–368. https://doi.org/10.1080/09692290.2020.1830832

Donnelly, S. (2018). *Power politics, Banking Union and EMU: Adjusting Europe to Germany*. Routledge.
Donnelly, S. (2023). Post-Brexit financial services in the EU. *Journal of European Public Policy*, *30*(5), 787–806. https://doi.org/10.1080/13501763.2022.2061579
European Banking Authority. (2014). EBA Opinion on 'virtual currencies'. (EBA/Op/2014/08).
European Banking Authority. (2018). The EBA's fintech roadmap. Conclusions from the consultation on the EBA's approach to financial technology (Fintech). https://bit.ly/3EUUucn.
European Central Bank. (2020). *ESCB/European banking supervision response to the European Commission's public consultation on a new digital finance strategy for Europe/FinTech action plan*. https://bit.ly/3JZiE8Z.
European Central Bank. (2021). *ECB Annual Report on supervisory activities* 2021. https://bit.ly/3YlVF6b.
European Commission. (2018). *FinTech Action Plan: For a more competitive and innovative European financial sector*. (COM(2018) 109 final).
European Commission. (2020a). *Digital finance strategy for the EU. (COM(2020) 591 final)*.
European Commission. (2020b). *Proposal for a Regulation of the European parliament and of the European Council on a pilot regime for market infrastructures based on distributed ledger technology*. (COM/2020/594).
European Commission. (2020c). *Proposal for a Regulation on digital operational resilience for the financial sector*. (COM/2020/595 final).
European Commission. (2020d). *Proposal for a Regulation on Markets in Crypto-assets*. (COM(2020) 593 final).
European Commission. (2023). Joint Communication to the European Parliament, the European Council and the Council on "European Economci Security Strategy". https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52023JC0020.
European Insurance and Occupational Pensions Authority. (2021). *Discussion paper on blockchain and smart contracts in insurance*. (EI-01-21-124-EN-N).
European Parliament. (2020a). *Verbatim report of proceedings*. Procedure 2020/2036 (INI). https://www.europarl.europa.eu/doceo/document/CRE-9-2020-10-07-ITM-015_EN.html.
European Parliament. (2020b). Digital Finance: emerging risks in crypto-assets - regulatory and supervisory challenges in the area of financial services, institutions and markets. https://www.europarl.europa.eu/doceo/document/TA-9-2020-0265_EN.html.
European Parliament. (2021a). *Draft Report of the ECON Committee on a pilot regime for market infrastructures based on distributed ledger technology*. (2020/0267(COD))). https://www.europarl.europa.eu/doceo/document/ECON-PR-689571_EN.pdf.
European Parliament. (2021b). *Draft Report on the proposal for a regulation of the European Parliament and of the Council on digital operational resilience for the financial sector*. (2020/0266(COD)). https://bit.ly/31ea8LI.
European Parliament. (2022). EP position at first reeding with amendments on the proposal for a regulation on markets in crypto-assets. (Document P9_A(2022)0052). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=EP%3AP9_A%282022%290052.
European Supervisory Authorities. (2019). *Joint Advice on the need for legislative improvements relating to ICT risk management requirements in the EU financial sector*. (JC 2019 26). https://bit.ly/3ZN62r7.

European Supervisory Authorities. (2022). *EU financial regulators warn consumers on the risks of crypto-assets*. (ESA, 2022 15).

Falkner, G., Heidebrecht, S., Obendiek, A., & Seidl, T. (2023). Digital sovereignty – rhetoric and reality: special issue framework paper. *Journal of European Public Policy*, Forthcoming.

Financial Stability Board. (2020). Regulation, Supervision and Oversight of 'Global Stablecoin' Arrangements. Final Report and High-Level Recommendations. https://www.fsb.org/wp-content/uploads/P131020-3.pdf.

Financial Stability Board. (2022). Assessment of risks to financial stability from crypto-assets. *Financial Stability Board Report*. https://bit.ly/3F81oKU.

Floridi, L. (2020). The fight for digital sovereignty: What It Is, and Why It matters, especially for the EU. *Philosophy & Technology*, *33*(3), 369–378. https://doi.org/10.1007/s13347-020-00423-6

Gabor, D., & Brooks, S. (2017). The digital revolution in financial inclusion: international development in the fintech era. *New Political Economy*, *22*(4), 423–436. https://doi.org/10.1080/13563467.2017.1259298

Geddes, B. (2003). *Paradigms and sand castles: Theory building and research design in comparative politics*. University of Michigan Press.

Heidebrecht, S. (2023). From market liberalism to public intervention: Digital sovereignty and changing European union digital single market governance. *JCMS: Journal of Common Market Studies*, 1–19. https://doi.org/10.1111/jcms.13488

Hesse, Jan-Otmar. (2021). Financial crisis and the recurrence of economic nationalism. *Journal of Modern European History*, *19*(1), 14–18. https://doi.org/10.1177/1611894420974254

Högenauer, A.-L., Howarth, D., & Quaglia, L. (2023). Introduction to the special issue: the persistent challenges to European Banking Union. *Journal of European Integration*, *45*(1), 1–14. https://doi.org/10.1080/07036337.2023.2183390

Howarth, D. J., & Quaglia, L. (2016). *The political economy of European banking union* (First ed.). Oxford University Press.

International Association for Trusted Blockchain Applications. (2021). INATBA Policy Position on Proposed Market in Crypto-Assets Regulation (MiCA).

Jabko, N. (2006). *Playing the market: A political strategy for uniting Europe, 1985–2005*. Cornell University Press.

Lambach, D., & Oppermann, K. (2022). Narratives of digital sovereignty in German political discourse. *Governance*, *36*(3), 693–709. https://doi.org/10.1111/gove.12690

Langley, P., & Leyshon, A. (2021). The platform political economy of FinTech: Reintermediation, consolidation and capitalisation. *New Political Economy*, *26*(3), 376–388. https://doi.org/10.1080/13563467.2020.1766432

Liebetrau, T. (2023). Problematising EU cybersecurity: Exploring How the single market functions as a security practice. *JCMS: Journal of Common Market Studies*, https://doi.org/10.1111/jcms.13523

Mansell, R. (2011). New visions, old practices: Policy and regulation in the Internet era. *Continuum*, *25*(1), 19–32. https://doi.org/10.1080/10304312.2011.538369

Morse, A. (2022). Meta's novi service to Be phased Out: What You need to know. *CNET*. https://www.cnet.com/personal-finance/crypto/metas-novi-service-to-be-phased-out-what-you-need-to-know/.

Obendiek, A. (2021). *Take back control? Digitale Souveränität und Europas digitale Zukunft*. *Hertie School Jaques Delors Centre Policy Paper*.

Pierson, P. (2000). Increasing returns, path dependence, and the study of politics. *American Political Science Review*, *94*(2), 251–267. https://doi.org/10.2307/2586011

Pohle, J., & Thiel, T. (2020). Digital sovereignty. *Internet Policy Review*, *9*(4), 1–19.
Posner, E., & Quaglia, L. (2023). Financial regulatory conundrums in the North Atlantic. *European Journal of Political Research*, 1–23. https://doi.org/10.1111/1475-6765.12616.
Quaglia, L., & Spendzharova, A. (2022). Disentangling derivatives: international policy reforms concerning central counterparties. *Journal of Economic Policy Reform*, *25*(3), 305–321. https://doi.org/10.1080/17487870.2021.1942874
Ríos Camacho, E. (2021). *The choice for banking union* London: Routledge.
Roberts, H., Cowls, J., Casolari, F., Morley, J., Taddeo, M., & Floridi, L. (2021). Safeguarding European values with digital sovereignty: an analysis of statements and policies. *Internet Policy Review*, *10*(3), 1–26. https://doi.org/10.14763/2021.3.1575.
Schmidt, V. A. (2011). Speaking of change: why discourse is key to the dynamics of policy transformation. *Critical Policy Studies*, *5*(2), 106–126. https://doi.org/10.1080/19460171.2011.576520
Schmitz, L., & Seidl, T. (2022). As open as possible, as autonomous as necessary: Understanding the rise of open strategic autonomy in EU trade policy. *JCMS: Journal of Common Market Studies*, *61*(3), 834–852.
Seawright, J., & Gerring, J. (2008). Case selection techniques in case study research: A menu of qualitative and quantitative options. *Political Research Quarterly*, *61*(2), 294–308. https://doi.org/10.1177/1065912907313077
Seidl, T., & Schmitz, L. (2023). Moving on to not fall behind? Technological sovereignty and the 'geo-dirigiste' turn in EU industrial policy. *Journal of European Public Policy*, 1–28. https://doi.org/10.1080/13501763.2023.2248204
Smith-Meyer, B. (2022). How Diem became crypto's sacrificial lamb *Politico*. https://www.politico.eu/article/diem-crypto-sacrificial-lamb/.
Streeck, W., & Thelen, K. A. (2005). *Beyond continuity: Institutional change in advanced political economies*. Oxford University Press.
Verdun, A. (2015). A historical institutionalist explanation of the EU's responses to the euro area financial crisis. *Journal of European Public Policy*, *22*(2), 219–237. https://doi.org/10.1080/13501763.2014.994023
von der Leyen, U. (2019). Mission letter to Valdis Dombrovskis executive vice-president for an economy that works for people.

Appendix 1: List of Interviews.

No.	Position	Date	Location
I	Senior Adviser, European Parliament	23. March 2022	Brussels
II	Former Member of European Parliament	25 March 2022	Telephone
III	Senior Adviser European Parliament	25 March 2022	Brussels
IV	Senior advisor, political party	25 March 2022	Brussels
V	Senior Advisor, Private Business and German think tank	24. May 2022	Zoom
VI	Senior Policy Advisor European Commission	16 February 2022	Zoom